# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: Artificial Intelligence** **COURSE CODE: CSA17**

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

*Assessment Tool Weightage: 35%*

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**QUESTIONS: 5** **Duration : 1 Hour**
**Total Marks:50**

Annexure A

**Analytical Problem Solving**

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***Code of Conduct:***

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

Annexure A

# Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.

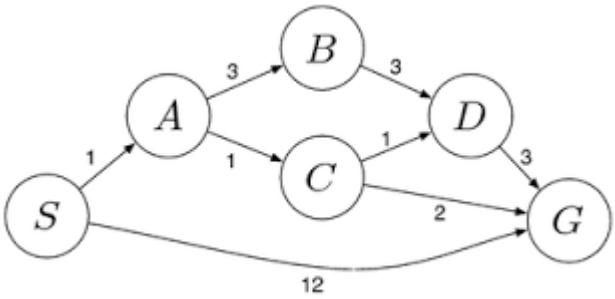
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
   i. What are the percept's for this agent?
   ii. Characterize the operating environment.
   iii. What are the actions the agent can take?
   iv. How can one evaluate the performance of the agent?
   v. What sort of agent architecture do you think is most suitable for this agent? Why?

3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.

4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.

5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse **S** to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

   The delivery network is represented as a weighted graph:




**RUBRICS FOR CALCULATION:**

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection | 20% | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| Accuracy of Calculation | 20% | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |

# Answer 1 – Water Jug Problem

Given:
Jug A = 4 gallons
Jug B = 3 gallons
Goal = Get exactly 2 gallons in the 4-gallon jug.

Steps:
1. Start → (0,0)
2. Fill 4G jug → (4,0)
3. Pour 4G → 3G → (1,3)
4. Empty 3G jug → (1,0)
5. Pour 1G → 3G → (0,1)
6. Fill 4G jug → (4,1)
7. Pour 4G → 3G until 3G is full → (2,3)

Answer: Exactly 2 gallons remain in the 4-gallon jug.

# Answer 2 – Mars Rover – Intelligent Agent

i) Percepts:
Camera images; soil and rock information; temperature; position/location; battery level; sample analysis results.

ii) Operating Environment:
Partially observable, uncertain, sequential, dynamic, continuous, and mostly unknown.

iii) Actions:
Move and turn; avoid obstacles; collect samples; analyse samples; take photographs; store/transmit data; manage battery power.

iv) Performance Evaluation:
Area explored; scientific information collected; quality of samples; safety; energy efficiency; successful completion of missions.

v) Suitable Agent Architecture:
Model-based goal-based agent.

Reason: The rover cannot observe everything directly. It maintains an internal model of its surroundings and chooses actions to achieve goals such as exploration and sample analysis.

# Answer 3 – 8-Queens Problem

Problem: Place 8 queens on an 8 × 8 chessboard so that no two queens attack each other.

Problem formulation:
• Initial state: Empty chessboard.
• State: Positions of queens already placed.
• Actions: Place a queen in a safe position in the next column.
• Transition: Add the selected queen position.
• Goal test: All 8 queens are placed and no two queens share the same row, column, or diagonal.
• Path cost: Each queen placement can have a cost of 1.

One valid solution:
[1, 5, 8, 6, 3, 7, 2, 4]

This represents the row position of the queen in each column.

Suitable search: Depth-First Search with backtracking.

# Answer 4 – OLA Cab Booking

Type of agent: Utility-based problem-solving agent.

Reason: The customer can select a cab based on cost, travel time, comfort, availability, and cab type. The agent compares these choices and selects the most suitable option.

Pseudocode:

```
START
Input source and destination
Get available cab types

For each available cab:
Calculate estimated fare
Calculate estimated travel time
Check comfort and availability
Calculate utility

Select cab with highest utility
Display selected cab and fare
Confirm booking
END
```

# Answer 5 – Uniform Cost Search (UCS)

Given: A weighted delivery network where each route has a cost.

UCS always expands the node having the lowest cumulative path cost.

Steps:
1. Start from S with cost 0.
2. From S: S → A = 1; S → G = 12.
3. Select A because its cost is 1.
4. From A: A → C = 1, so cost = 2; A → B = 3, so cost = 4.
5. Select C because its cost is 2.
6. From C: C → D = 1, so cost = 3; C → G = 2, so cost = 4.
7. Select G with cost 4.
8. Goal reached.

Optimal path:
S → A → C → G

Total cost:
1 + 1 + 2 = 4

Final Answer:
Least-cost path = S → A → C → G
Minimum cost = 4